

Network Programming

BCA — Semester VI — 2025 — Answer Sheet

Subject Code: CACS355

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Describe the scope and applications of Java Socket and Node.js in network programming platforms.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the purpose of the NetworkInterface class and describe its factory and getter methods with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain the process of opening a URL connection and reading data from the server in Java.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write a Program to Send and Receive a Message using Connectionless Sockets.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What are the fundamental components of JAVA NIO? Difference between Blocking and Non-blocking I/O.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain the RMI architecture with the detail role.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes. (Any two)a) ProxySelectorb) JSSEc) Web Cache

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. a) What is Cookie? Explain about CookieManager and HttpCookie classes. b) Discuss the purpose and use of HTTP response codes in URL connections. Provide examples of common response codes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain how a TCP echo server using thread created with snipped code and also give their advantages.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is Traceroute? Write down the working principle. Explain the Trace Route program with sample code.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.